

BrandGuard AI Capstone Project Report

AI-Powered Brand-Safe Marketing Content Generation System

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Name of the project- Brand Safe Content and campaign studio.

Abstract

BrandGuard AI is a workflow automation solution built with n8n, OpenAI and Google Sheets. It automates marketing content generation using approved product information and brand guidelines while validating outputs and supporting human review.

Introduction

This project demonstrates the integration of Generative AI with workflow automation to create brand-compliant marketing content efficiently.

Problem Statement

Organizations invest considerable time and effort in producing marketing content that complies with internal brand policies. Manual review processes are slow, repetitive, and susceptible to human error. There is a need for an intelligent system that generates high-quality marketing content while ensuring compliance with approved product information and organizational branding standards before publication.

Objectives

- * Develop an AI-powered content generation workflow.
- * Read approved product information from Google Sheets.
- * Read company brand guidelines.
- * Generate professional marketing content using OpenAI.

- * Validate that the generated content is not empty.
- * Store generated content automatically.
- * Support human review before publishing.
- * Demonstrate workflow automation using n8n.

Technology Stack

n8n Cloud

OpenAI API

Google Sheets

AI Agent

IF Node

System Architecture

User Starts Workflow

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Manual Trigger

|



Google Sheets

(Product Database)

|



Google Sheets

(Brand Guidelines)

|



AI Agent

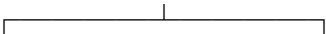
(OpenAI)

|



Validation (IF Node)

|



| |

True False

| |

Append Row Stop Workflow

|



Generated_Content Sheet

|



Pending Human Approval

Implementation

Describe each node:

1. Manual Trigger
2. Google Sheets (Products)
3. Google Sheets (Brand Guidelines)
4. AI Agent
5. OpenAI Chat Model
6. IF Validation
7. Google Sheets Append Row

[Insert workflow screenshots after each subsection.]

Testing

Test Case 1

Input:

SmartWatch X

Result:

Instagram marketing content successfully generated.

Validation:

Passed

Status:

Pending

Results

The workflow successfully generated marketing content, validated responses and stored outputs with Pending approval.

Future Scope

Email approvals, Slack integration, CRM integration, multilingual support and analytics dashboard.

Conclusion

The project demonstrates responsible AI through Human-in-the-Loop approval while reducing manual effort and improving consistency.

Screenshot Placeholders

Figure 1: Complete n8n Workflow

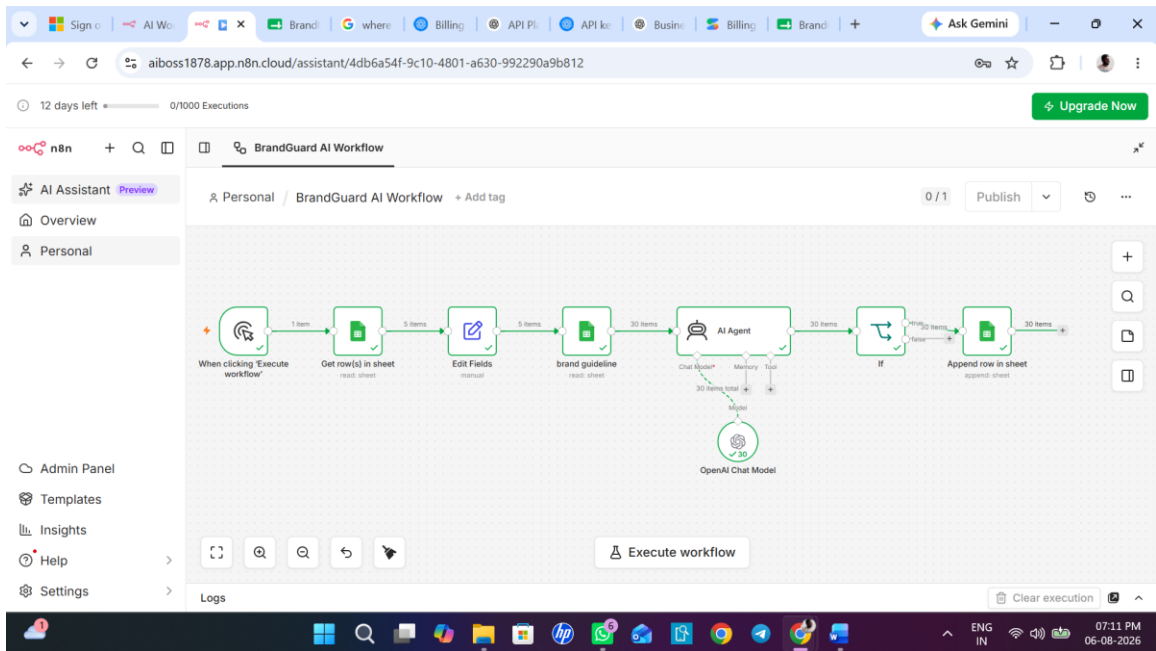


Figure 2: AI Agent Configuration

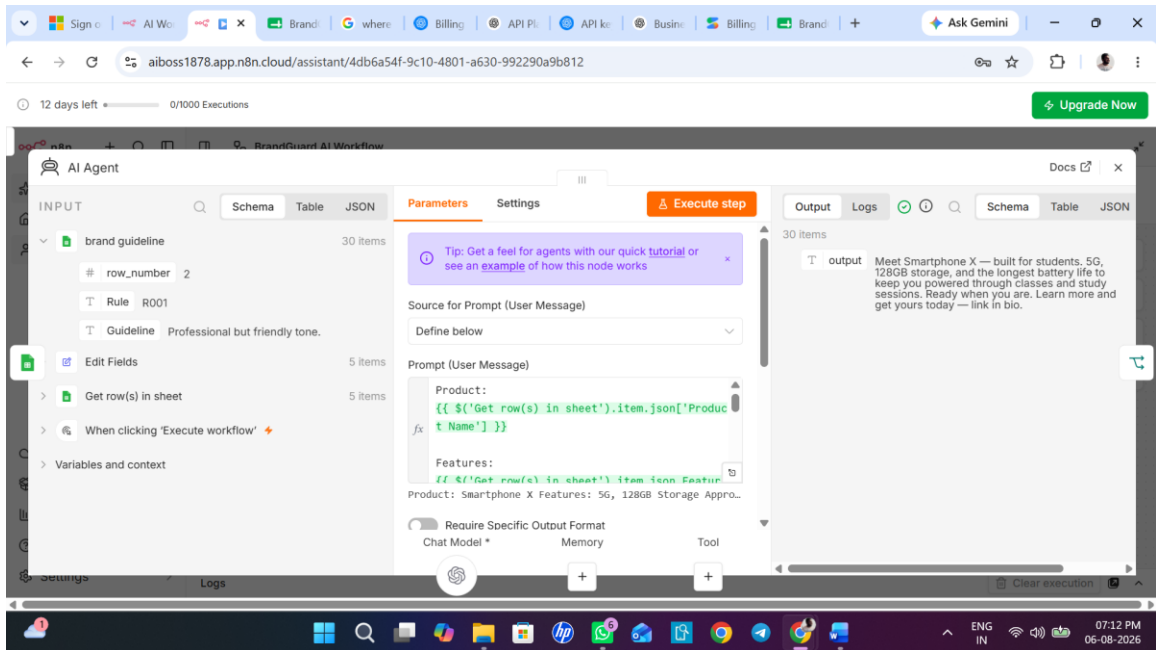


Figure 3: IF Node

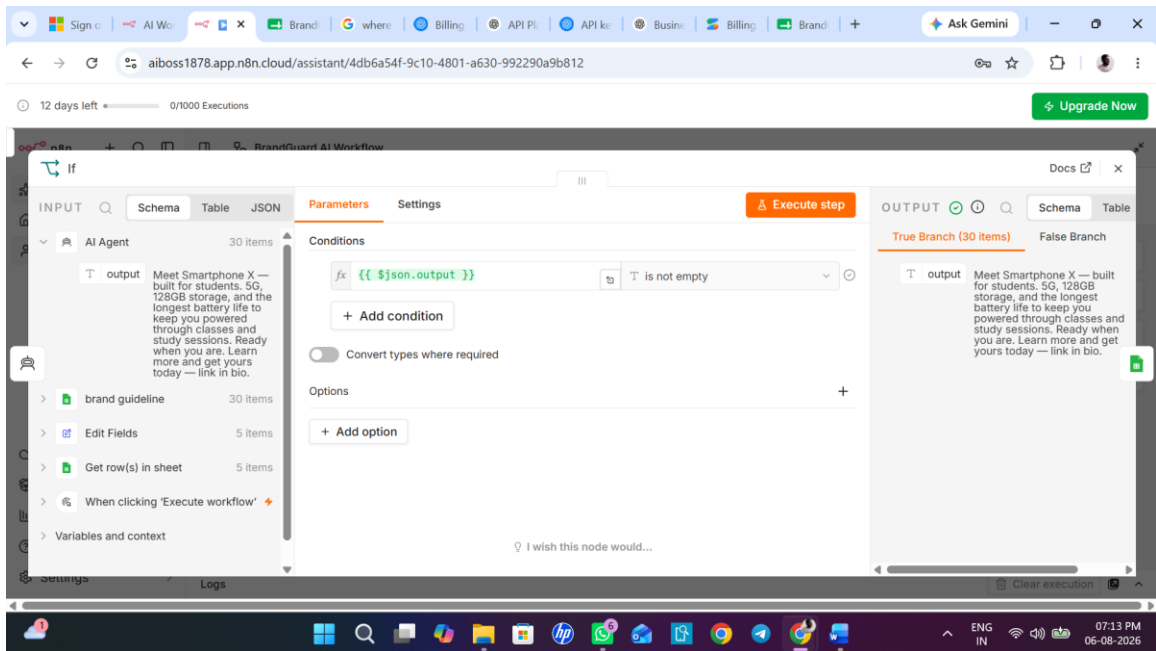


Figure 4: Generated_Content Google Sheet

	A	B	C	D	E	F	G	H	I	J	K	L
1	Date	Product	Content type	AI Output	Validation	Approval status	output					
2	2026-08-06T18:18:18	Smartphone X	Instagram Post	Meet Smartphone X — built for students. 5G, 128GB storage, and the longest battery life to keep you powered through classes and study sessions. Ready when you are. Learn more and get yours today — link in bio.	Passed	Pending	Meet Smartphone X — built for students. 5G, 128GB storage, and the longest battery life to keep you powered through classes and study sessions. Ready when you are. Learn more and get yours today — link in bio.					
3	2026-08-06T18:18:18	Smartphone X	Instagram Post	Smartphone X — built for students. 5G, 128GB storage, and the longest battery life to keep you powered through classes and study sessions. Ready when you are. Learn more and get yours today — link in bio.	Passed	Pending	Smartphone X — built for students. 5G, 128GB storage, and the longest battery life to keep you powered through classes and study sessions. Ready when you are. Learn more and get yours today — link in bio.					
4	2026-08-06T18:18:18	Smartphone X	Instagram Post	Students: Smart	Passed	Pending	Students: Smartphone X — 5G • 128GB • Longest battery life. Ready to upgrade? Tap to learn more and get yours today — link in bio.					
5	2026-08-06T18:18:18	Smartphone X	Instagram Post	Students: Smart	Passed	Pending	Students: Smartphone X offers 5G connectivity, 128GB storage, and the longest battery life — so you stay focused during lectures and study with Smartphone X.					
6	2026-08-06T18:18:18	Smartphone X	Instagram Post	Students: meet	Passed	Pending	Students: meet Smartphone X — 5G, 128GB storage, and the longest battery life. Built for lectures, study sessions, and streaming. Ready when you are. Learn more and get yours today — link in bio.					
7	2026-08-06T18:18:18	Smartphone X	Instagram Post	Meet Smartphone X	Passed	Pending	Meet Smartphone X — built for students who do more. With 5G, 128GB storage and the longest battery life, Smartphone X is the perfect companion for your studies.					
8	2026-08-06T18:18:18	Wireless Earbud	Instagram Post	Wireless Earbud	Passed	Pending	Wireless Earbuds Pro — Active Noise Cancellation and Crystal clear sound. Ideal for students during study sessions and lectures. Ready when you are. Learn more and get yours today — link in bio.					
9	2026-08-06T18:18:18	Wireless Earbud	Instagram Post	Wireless Earbud	Passed	Pending	Wireless Earbuds Pro — includes Active Noise Cancellation and delivers crystal clear sound. Designed for students who want to stay focused during lectures and study.					
10	2026-08-06T18:18:18	Wireless Earbud	Instagram Post	Students: meet	Passed	Pending	Students: meet Wireless Earbuds Pro — featuring Active Noise Cancellation and Crystal clear sound. Designed for students who want to stay focused during lectures and study.					
11	2026-08-06T18:18:18	Wireless Earbud	Instagram Post	Students: focus	Passed	Pending	Students: focus during lectures and study with Wireless Earbuds Pro. Active Noise Cancellation reduces distractions, so you can stay focused on your studies.					
12	2026-08-06T18:18:18	Wireless Earbud	Instagram Post	Students — stay	Passed	Pending	Students — stay focused with Wireless Earbuds Pro. Active Noise Cancellation reduces distractions, so you can stay focused on your studies.					
13	2026-08-06T18:18:18	Wireless Earbud	Instagram Post	Students: stay fo	Passed	Pending	Students: stay focused with Wireless Earbuds Pro. Active Noise Cancellation minimizes distractions, so you can stay focused on your studies.					
14	2026-08-06T18:18:18	4K Smart TV 55"	Instagram Post	Students: Bring	Passed	Pending	Students: Bring big-screen clarity to your study and downtime with the 4K Smart TV 55". Enjoy 4K resolution, Smart OS, and vibrant colors.					
15	2026-08-06T18:18:18	4K Smart TV 55"	Instagram Post	Students — upgr	Passed	Pending	Students — upgrade your dorm with the 4K Smart TV 55". Vivid colors, 4K resolution, and a Smart OS make it the perfect addition to your study space.					
16	2026-08-06T18:18:18	4K Smart TV 55"	Instagram Post	Students: Discov	Passed	Pending	Students: Discover the 4K Smart TV 55". Features: 4K resolution, Smart OS, and vibrant colors. Want details? Visit our website today.					
17	2026-08-06T18:18:18	4K Smart TV 55"	Instagram Post	Upgrade your do	Passed	Pending	Upgrade your dorm with the 4K Smart TV 55". Enjoy vivid colors and true 4K resolution on a 55" screen. Ready when you are. Learn more and get yours today — link in bio.					

Access link to the project (Github) - <https://github.com/Omkar1878/BrandGuard-AI-Capstone>